

Translation Literacy in Asylum Interpreting: A Pilot Study of TRACE and Structured Reflection

Mohammad Shehadeh

Abstract

Interpreting quality can shape how an asylum applicant's account is understood, while UK inspection and guidance documents continue to identify challenges involving language matching, interpreter preparation and quality assurance. This pilot study operationalises translation literacy as explicit reflection on translation choices, semantic range, alternatives, evidence, consequences and evaluation. Twenty-one Arabic-English bilingual adults completed unguided pre- and post-framework translation-evaluation tasks after guided exposure either to TRACE, a five-step mnemonic reflection scaffold, or to an active structured control checklist. Eighty-four responses were scored on a six-dimension rubric by the original marker and three independent blinded raters; analyses used the mean of the four ratings. TRACE did not produce a statistically significant advantage over control on combined change scores (TRACE +1.48/24; control +2.33/24; $U=47.0$, $p=.597$). Alternatives showed the clearest TRACE-aligned descriptive pattern, but the between-group change difference was non-significant. The findings therefore do not establish TRACE superiority or a general causal effect of structured reflection. They instead identify a feasible measurement approach and hypotheses for larger, better-balanced studies using professional interpreters, delayed retention testing and spoken high-stakes tasks.

Keywords: asylum interpreting; interpreter education; metacognition; public-service interpreting; translation competence

1. Introduction

In UK asylum procedures, interpreting is not ancillary to the interview: it is the medium through which many applicants present the account on which protection decisions are made. Paragraph 339ND of the Immigration Rules requires the Secretary of State to provide an interpreter where necessary and to select one who can ensure appropriate communication between the applicant and the interviewing representative (Home Office, n.d.). The current asylum-interview guidance adds operational expectations. Interpreters are expected to confirm the correct language or dialect, retain every element of the original message, avoid summarising, request clarification when necessary, and, where a direct equivalent is difficult, use the closest equivalent while informing the interviewer (Home Office, 2025). These requirements make translation decisions consequential, but they do not by themselves explain how an interpreter should examine uncertainty or compare plausible alternatives while developing professional judgement.

Concerns about competence and preparation have also entered public debate. In 2025, Baroness Coussins proposed an amendment that would have replaced the Immigration Rules' reference to 'an interpreter' with 'a qualified, professional interpreter'; the amendment was withdrawn after debate (UK Parliament, 2025). The proposal is not evidence that formal qualification alone resolves interpreting quality. It does, however, illustrate a persistent question for high-stakes public-service interpreting: what forms of competence should be expected, how should they be taught, and how can they be assessed?

This paper examines one deliberately narrow part of that question. It introduces TRACE, a mnemonic scaffold intended to make translation reasoning more explicit, and reports a small

pilot comparison with an active structured control. The study does not evaluate real asylum interviews, professional accreditation or the effectiveness of Home Office policy. Instead, it asks whether a short reflective framework changes the kinds of reasoning that Arabic-English bilingual adults display in controlled translation-evaluation tasks, and whether any TRACE-specific pattern remains when the prompts are removed.

2. Background and literature review

2.1 Asylum interpreting as a high-stakes language task

UK evidence points to two distinct levels of difficulty that should not be conflated. Some problems are structural: whether an interpreter in the required language or dialect is available, how interpreters are procured, and what training or quality-assurance arrangements surround their work. Other problems occur inside the interpreted interaction, where choices about wording, register, semantic range and clarification affect the communicative record. TRACE addresses only the second level, and even there only as a possible educational scaffold rather than as a remedy for systemic problems.

The Independent Chief Inspector of Borders and Immigration (ICIBI) examined Home Office language services in 2020, including the quality and accuracy of interpreting and the mechanisms used for quality assurance. The inspection described a 'rare and difficult' route for languages in which recognised qualifications were less readily available and reported that interpreters received guidance through a Code of Conduct rather than a dedicated Home Office training programme (ICIBI, 2020). Williamson's (2024) study similarly characterises UK public-service interpreting as highly deregulated and argues that asylum work involves complex linguistic and sociocultural mediation rather than mechanical language transfer. These sources support the need to examine preparation and reflective competence, but they do not demonstrate that a particular reflective framework improves interpreting.

More recent inspection evidence identified practical concerns in substantive asylum interviews, including some interpreters' English proficiency, interpreting quality and access to the appropriate language or dialect (ICIBI, 2024). Dialect mismatch is especially relevant to Arabic, where regional varieties can differ substantially in vocabulary, phonology and pragmatic convention. Current Home Office guidance consequently requires the claimant and interpreter to understand one another and directs the interpreter to raise differences in language or dialect at the earliest opportunity (Home Office, 2025). A reflective framework can make attention to such questions explicit in training; it cannot create an unavailable dialect match or substitute for competent recruitment.

The emotional conditions of refugee-related interpreting provide a separate challenge. Morrison, Given-Wilson and Memon (2024) studied 28 Arabic-English interpreters in a mock asylum interview and found that emotionally evocative responses were interpreted 4%-8% less accurately than neutral responses, a statistically significant medium-to-large effect. Geiling et al. (2022b), in a survey of 164 interpreters working in refugee care in Germany, found elevated psychological distress and associations between work-related exhaustion and factors including dissatisfaction with payment and exposure to traumatic content. A related study of 158 interpreters found differences in wellbeing and working conditions across psychotherapy, counselling, medical and authority settings (Geiling et al., 2022a). These studies show that refugee-related interpreting can take place under demanding conditions; they do not show that

distress directly caused errors in the present study, whose tasks were written, constructed and non-traumatic.

2.2 Translation competence, metacognition and explicit reasoning

The concept of translation literacy is used here as an operational construct rather than as an established psychometric scale. It refers to a participant's ability to notice a translation problem or uncertainty, identify the semantic range at stake, generate or compare alternatives, connect claims to textual evidence, consider the consequences of wording choices, and reach an evaluative judgement. This definition is intentionally process-oriented: it concerns the reasoning that can be articulated around a translation choice, not a claim that every choice has a single objectively correct answer.

This emphasis is compatible with broader models of translation competence. PACTE's empirical work treats translation competence as involving strategic decision-taking and problem solving alongside bilingual and extra-linguistic resources (PACTE, 2005). Baker's (2018) discussion of non-equivalence at word level shows why translators often face choices among partial rather than perfect correspondences. Borg (2019) demonstrates that alternative translation solutions are a recurrent part of translatorial decision-making. Together, these perspectives motivate an educational approach that asks learners not only what they would translate, but what alternatives they considered and why one option is preferable in context.

Metacognitive research offers a more direct precedent. Hu, Zheng and Wang (2021) developed a metacognitive self-regulation inventory organised around planning, monitoring and evaluation. In an English-Chinese student pre-post study, repeated use of the inventory was associated with greater metacognitive awareness and improvements in aspects of translation quality, including clarity, vocabulary, morphosyntax, genre conventions, purpose and target audience. Their study suggests that structured reflection can be trainable, but its intervention, population and outcomes differ from those used here. The present pilot therefore treats Hu et al. as a rationale for testing reflection, not as evidence that TRACE should work.

A further issue is measurement. Rater-mediated assessment of interpreting and translation requires human judgement, and analytic rubrics do not eliminate differences in how raters attend to source material, descriptors or error types. Han et al. (2024), using eye tracking and retrospective verbalisation with 35 raters, showed that interpreting assessment involves multiple scoring strategies and complex interaction with an analytic scale. This is relevant to the present pilot because its six-dimension rubric was itself exploratory. A useful intervention cannot be inferred from a measure whose reliability and validity are assumed rather than examined.

2.3 Professional qualification and the scope of TRACE

The CIOL Qualifications Level 6 Diploma in Public Service Interpreting (DPSI) is an Ofqual-regulated, nationally recognised qualification that CIOL describes as a benchmark for public-service interpreting. Its five units assess consecutive and simultaneous interpreting, sight translation into and from English, and written translation into and from English (Chartered Institute of Linguists [CIOL], n.d.). These are professional performance outcomes. TRACE is narrower: it is designed to scaffold explicit reflection on decisions made during or after a translation task. The present study did not recruit DPSI candidates or holders as a defined sample and cannot compare TRACE with professional qualification.

This distinction matters for interpretation of the study. A qualification asks whether a candidate can perform required tasks to a professional standard; a reflective scaffold asks whether particular reasoning habits can be elicited, practised and perhaps retained. Those functions may eventually complement one another, but the current data address only the second. Any claim that TRACE improves professional interpreting, asylum-interview accuracy or DPSI performance would require direct testing in those settings.

3. The TRACE framework

3.1 Design principles

TRACE was designed as a compact mnemonic for five linked questions that can be applied to a translation already produced or to a translation decision in progress. It is not an accuracy algorithm and does not prescribe one correct wording. Rather, it makes uncertainty inspectable. The framework asks the user to establish the communicative situation, map the relevant semantic range, generate alternatives, consider their consequences and then justify an evaluation. In pedagogical terms, the scaffold is intended to move reflection from an unsupported judgement ('this sounds wrong') toward an evidenced comparison ('this option changes register or agency; this alternative preserves the source effect more closely').

Step	Function	Core question	Illustrative prompts	Rubric emphasis
T — Text and transmission	Situate the communicative act	What is being transmitted, by whom, to whom, and under what constraints?	What language/dialect or register is being used? What is the purpose and audience? Is anything about delivery relevant?	Specificity; evidence
R — Range of semantic meaning	Identify uncertainty and semantic possibilities	What might the relevant source expression mean here?	What senses, connotations or pragmatic readings are plausible? What does a target choice add or remove?	Range; evidence

Step	Function	Core question	Illustrative prompts	Rubric emphasis
A — Alternatives	Generate competing solutions	What else could reasonably be said?	Can I produce another translation? Does a more literal, idiomatic, formal or colloquial option preserve something different?	Alternatives
C — Consequences	Compare effects of the alternatives	Why would the choice matter?	Does it alter tone, certainty, agency, politeness, credibility, relationship or implied stance?	Consequences; evidence
E — Evaluation	Reach a justified and provisional judgement	Which option is preferable for this purpose, and how confident am I?	What does the chosen version preserve or lose? Are several versions defensible? What uncertainty remains?	Evaluation; evidence

Table 1. TRACE as a five-step reflective translation scaffold. The rubric emphasis column describes conceptual alignment; it is not a claim that each TRACE step maps uniquely to one scored dimension.

3.2 How the steps work together

The five letters are presented in sequence for memorability, but the reasoning is not strictly linear. A translator may identify an alternative and then return to Range after noticing a new implication; considering Consequences may reveal that the Text and transmission context was underspecified; Evaluation may remain provisional because two alternatives are defensible. TRACE therefore functions more like a short loop for disciplined reconsideration than a checklist that ends once five boxes have been ticked.

Text and transmission deliberately comes first because an apparently lexical problem can be contextual. In public-service interpreting, the relevant context may include dialect, speaker relationship, direct speech, institutional purpose and the need to preserve every substantive element of an account. Current Home Office guidance itself requires attention to correct language or dialect, completeness, clarification and closest equivalents (Home Office, 2025).

TRACE does not reproduce that guidance or create a new professional rule; it provides a pedagogical language for asking why a choice may or may not satisfy such communicative constraints.

Range and Alternatives separate two processes that are often collapsed. Range asks what the source could mean; Alternatives asks what the target could say. The distinction is useful because recognising ambiguity does not guarantee that a learner can generate a better rendering, while producing a different wording does not guarantee awareness of the source-language range. Consequences then requires comparison at the level of communicative effect rather than surface difference. Finally, Evaluation asks for a judgement that is both justified and revisable. A defensible outcome may be to retain the original translation.

The active control used in this study also promoted structured attention: participants were asked to read, clarify, check language, identify problems and give an overall judgement. The comparison is therefore deliberately demanding. If both conditions improve, this design cannot tell whether structure itself caused the change; if TRACE outperforms control, a more specific mechanism becomes plausible. The pilot was accordingly framed around comparative and transfer questions rather than a simple pre-post claim.

3.3 Research questions

The pilot addressed three research questions:

- RQ1. Does TRACE produce greater pre- to post-framework change than an active structured control on the six-dimension translation-literacy score?
- RQ2. Do TRACE participants show a distinctive descriptive pattern in Alternatives or in the breadth of dimensions used once the scaffold is removed?
- RQ3. Are the principal conclusions robust to four-rater aggregation and to a sensitivity analysis excluding responses with uncertain authorship?

4. Method

4.1 Ethics, recruitment and participants

All participants were adults recruited through Prolific. They received study information, provided informed consent, and could withdraw in accordance with the study procedure. Responses were anonymised, and participants separately indicated whether anonymised responses could be shared or quoted. The study used constructed translation materials rather than asylum case files or testimony and did not recruit asylum applicants.

Participants were pre-screened for Arabic and English fluency and were required to be at least 18 years old. The survey collected age, education, Arabic variety, translation experience, translation frequency and self-rated language and translation ability. These variables were used descriptively to characterise the sample and inspect possible baseline imbalance; the sample was too small for credible moderator modelling.

4.2 Design and materials

The survey used six constructed passages generated with ChatGPT 5.6 Sol: two Arabic social-media posts (versions A and B) paired with English translations for critique; two English diary entries (A and B) for translation into Arabic and self-review; one Arabic storybook passage with an English translation for guided critique; and one English historical-style passage for guided English-to-Arabic translation and self-review. Constructed material allowed tighter

control and avoided use of sensitive asylum records, but it reduced ecological validity. The tasks did not reproduce live speech, interactional turn-taking, time pressure, memory load or emotionally difficult testimony.

Participants were assigned using permuted blocks of four. Within each block, TRACE+A, TRACE+B, control+A and control+B were shuffled and issued sequentially to new participant IDs. Returning participants retained their original assignment. A shared Google Apps Script serialised assignments with a lock and recorded them; a browser-local fallback used the same four-combination scheme but could not guarantee balance across browsers. Despite the intended balance, the analysed sample contained five participants in Set A and sixteen in Set B. The retained records do not establish why the imbalance occurred, so passage-set analyses are descriptive and the failure of intended balancing is treated as an implementation limitation.

4.3 Conditions and procedure

Before framework exposure, participants completed two unguided baseline tasks: critique of a supplied English translation of an Arabic social-media post and translation of an English diary entry into Arabic followed by self-review. Version A participants saw Social A and Diary A at baseline; version B participants saw Social B and Diary B. Participants were then shown their allocated framework with guided practice using the same storybook and historical-style passages across A/B versions. In the post-framework assessment the scaffold was removed. Participants repeated the same task types using the opposite A/B passages. The design therefore assessed immediate unguided carry-over after framework exposure, not delayed retention.

The control checklist asked participants to:

- Read: summarise the text's main meaning.
- Clarify: decide whether the translation is clear and understandable.
- Language: check vocabulary, grammar and naturalness.
- Problems: identify any problem or uncertainty.
- Judgement: give and explain an overall judgement.

4.4 Rubric, independent marking and score aggregation

Each of the 84 assessment responses was scored on six dimensions from 0 to 2: Specificity, Range, Alternatives, Evidence, Consequences and Evaluation. A score of 0 indicated absence of the relevant feature; 1 indicated a limited, vague or singly supported demonstration; and 2 indicated a clear and justified demonstration, usually with multiple or well-developed examples. Detailed descriptors and examples are supplied in Supplementary Table S1.

After the original marking, three independent Arabic-English reviewers were recruited through Prolific. Each reviewer received all 84 responses in a blinded, randomised order and did not see participant IDs, condition, pre/post status, passage set, uncertain-authorship flags or original scores. Four ratings were therefore available for every response. For each dimension, the arithmetic mean of the four ratings was calculated and the six dimension means were summed to produce the /12 response score. Pre-specified use of the mean reduced dependence on one marker and avoided selecting an aggregation method after viewing the TRACE-control result.

Inter-rater reliability was evaluated before interpreting the consensus scores. A two-way random-effects absolute-agreement intraclass correlation gave $ICC(A,1)=.303$ for a single rater and $ICC(A,4)=.635$ for the mean of four raters. Fleiss' kappa for exact 0-2 category agreement ranged from $-.002$ to $.139$ across dimensions. These values indicate substantial rater variation at

category level and only moderate reliability for the averaged total, reinforcing the exploratory status of the rubric. This kind of rater variability is consistent with evidence that interpreting assessment involves complex interaction with analytic rating scales (Han et al., 2024).

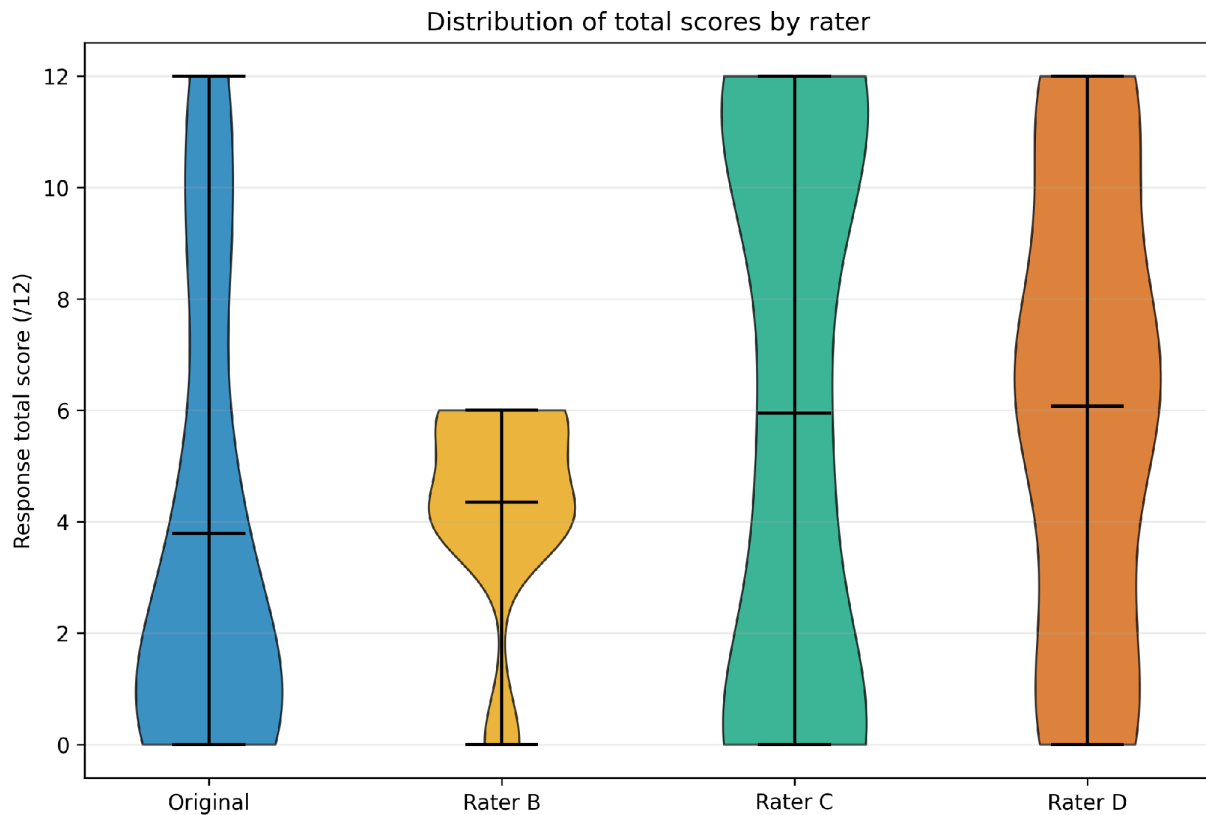


Figure 1. Distribution of total response scores (/12) by rater. Raters are anonymised as Original and Raters B-D.

4.5 Statistical analysis

Because the sample was small ($n=21$) and outcomes were bounded, inferential analyses were exploratory. Wilcoxon signed-rank tests assessed pre-post change within each condition, and Mann-Whitney U tests compared change scores between TRACE and control. Tests were two-sided and p-values were not adjusted for multiple comparisons. The primary descriptive outcome was the combined /24 score formed by adding the self-review and critique task scores within each phase. Secondary analyses examined individual rubric dimensions, participant-level direction of change and breadth, defined as the number of six dimensions with a mean four-rater score above zero, averaged across the two task types within a phase.

The primary analysis retained all participants. A sensitivity analysis excluded any participant whose response had been flagged for authorship uncertainty. The flags were based on marked within-participant discontinuities in style and elaboration, not on polished writing alone; no automated AI detector was treated as ground truth. The sensitivity analysis is therefore a robustness check for uncertain authorship, not a classification of proven AI use.

5. Results

5.1 Task-level outcomes

Across the full sample, mean self-review scores increased by +0.56/12 and critique scores by +1.32/12. TRACE and control showed similar self-review change (+0.61 and +0.50 respectively), while control showed the larger descriptive increase on critique (+1.83 versus +0.86). Neither between-group change difference was statistically significant (self-review $U=63.0$, $p=.597$; critique $U=42.0$, $p=.378$), and within-group tests were non-significant for both task types.

Task	Group	n	Pre mean (SD) [median]	Post mean (SD) [median]	Change mean (SD) [median]
Self-review	ALL	21	3.83 (2.75) [2.75]	4.39 (2.78) [3.25]	+0.56 (3.14) [0.25]
Self-review	TRACE	11	4.52 (3.45) [3.00]	5.14 (3.02) [3.25]	+0.61 (2.87) [0.50]
Self-review	Control	10	3.08 (1.51) [2.75]	3.58 (2.38) [2.75]	+0.50 (3.57) [-0.38]
Critique	ALL	21	5.30 (2.59) [5.00]	6.62 (2.39) [7.25]	+1.32 (2.59) [1.25]
Critique	TRACE	11	6.07 (2.92) [5.50]	6.93 (2.74) [8.00]	+0.86 (2.13) [1.00]
Critique	Control	10	4.45 (1.98) [4.50]	6.28 (2.04) [6.88]	+1.83 (3.05) [2.50]

Table 2. Full-sample task-level scores using the four-rater mean.

5.2 Rubric dimensions and immediate unguided carry-over

All six dimensions increased descriptively in both conditions. Control showed larger mean gains on Specificity, Range, Evidence, Consequences and Evaluation. Alternatives was the sole dimension with a larger TRACE change (+0.19 versus +0.14). This difference was not statistically significant ($U=56.5$, $p=.943$). TRACE also had a higher post-framework Alternatives mean (0.92 versus 0.63), but the post-only difference was non-significant ($U=78.0$, $p=.110$) and TRACE had already begun higher at baseline (0.73 versus 0.49). Alternatives is therefore a mechanism-aligned descriptive signal, not evidence of a TRACE-specific effect.

Group	Dimension	Pre	Post	Change
TRACE	Specificity	1.03	1.14	+0.10
TRACE	Range	0.97	1.05	+0.08
TRACE	Alternatives	0.73	0.92	+0.19

Group	Dimension	Pre	Post	Change
TRACE	Evidence	0.82	0.92	+0.10
TRACE	Consequences	0.86	1.01	+0.15
TRACE	Evaluation	0.89	1.00	+0.11
Control	Specificity	0.70	0.91	+0.21
Control	Range	0.71	0.90	+0.19
Control	Alternatives	0.49	0.63	+0.14
Control	Evidence	0.44	0.71	+0.28
Control	Consequences	0.70	0.89	+0.19
Control	Evaluation	0.73	0.89	+0.16

Table 3. Full-sample mean dimension scores before and after framework exposure.

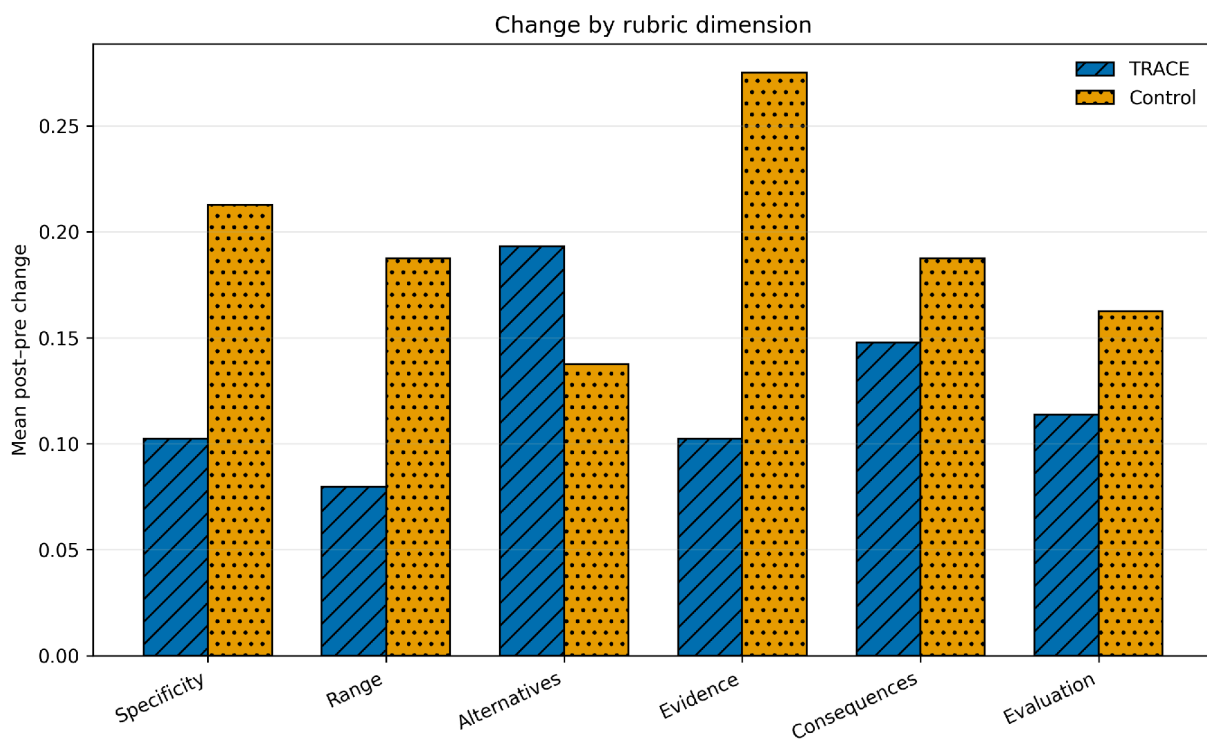


Figure 2. Mean post- minus pre-framework change by rubric dimension. Positive values indicate improvement.

5.3 Breadth, combined scores and participant-level change

Breadth was already near its ceiling. TRACE increased from 5.36 to 5.95 dimensions (+0.59) and control from 5.40 to 5.70 (+0.30); the between-group difference was non-significant ($p=.410$), and median change was zero. Because a dimension counted as present whenever its

mean score was above zero, this measure was sensitive to very modest evidence of a behaviour and had limited room to detect improvement.

On the combined /24 outcome, the full sample increased from 9.13 to 11.01 (+1.88). TRACE began at 10.59 and increased to 12.07 (+1.48), whereas control began at 7.53 and increased to 9.85 (+2.33). The between-group comparison was non-significant ($U=47.0$, $p=.597$). The result does not establish equivalence, but it does not support TRACE superiority in this sample.

Analysis	Group	n	Pre /24	Post /24	Change
Full sample	ALL	21	9.13 (4.66) [8.25]	11.01 (4.38) [10.00]	+1.88 (4.64) [1.75]
Full sample	TRACE	11	10.59 (5.57) [11.50]	12.07 (4.86) [11.25]	+1.48 (4.03) [0.75]
Full sample	Control	10	7.53 (2.88) [7.13]	9.85 (3.70) [8.75]	+2.33 (5.43) [3.25]
Uncertain-aut horship excluded	ALL	18	8.56 (4.61) [6.88]	10.72 (4.63) [9.50]	+2.17 (4.58) [1.88]
Uncertain-aut horship excluded	TRACE	9	9.75 (5.75) [7.00]	12.19 (5.42) [10.00]	+2.44 (3.66) [1.75]
Uncertain-aut horship excluded	Control	9	7.36 (3.00) [6.00]	9.25 (3.36) [8.75]	+1.89 (5.57) [2.50]

Table 4. Combined /24 scores in the full sample and uncertain-authorship sensitivity sample.

Participant-level direction of change was similarly mixed. Seven of eleven TRACE participants (63.6%) and seven of ten controls (70.0%) improved in the full sample; four TRACE and three control participants declined. After the sensitivity exclusions, 7/9 TRACE participants (77.8%) and 6/9 controls (66.7%) improved. The plot shows large positive and negative movements in both conditions, indicating heterogeneous responses rather than a uniform framework effect.

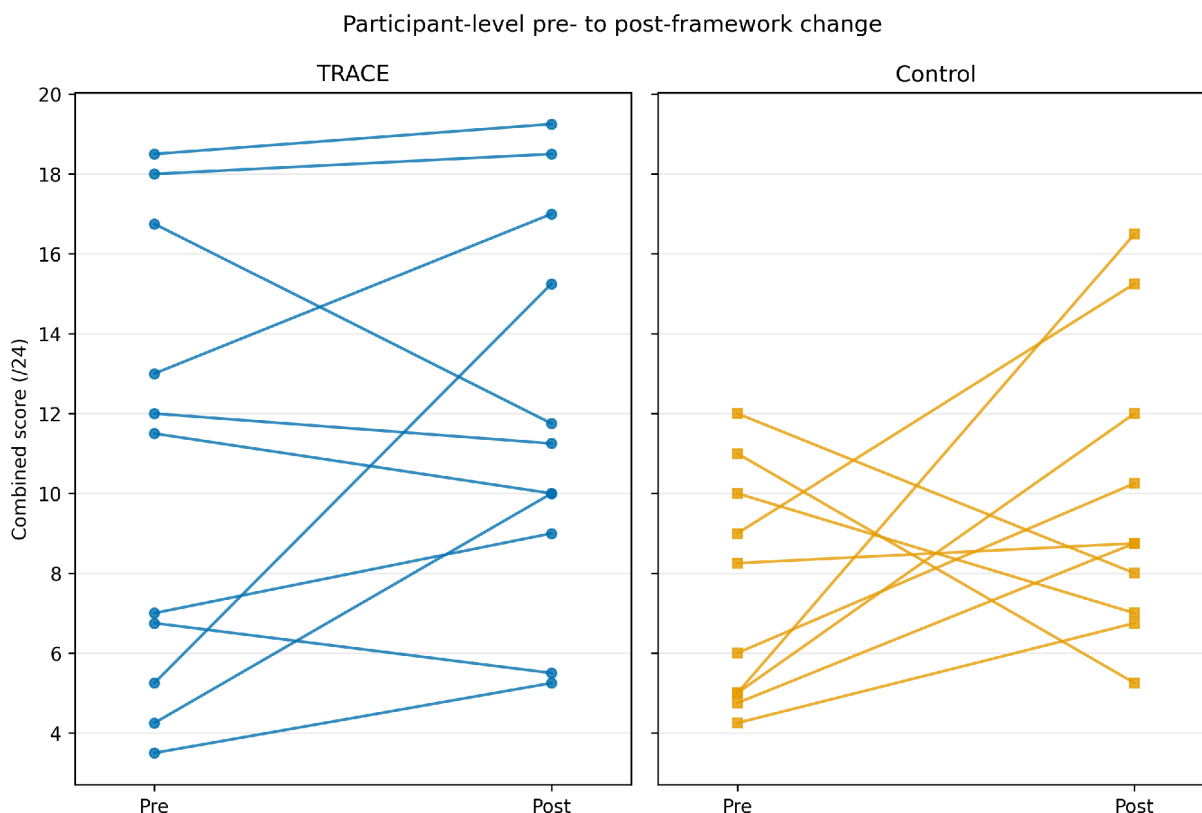


Figure 3. Participant-level pre- to post-framework change in the combined /24 score.

5.4 Passage-set imbalance

Passage version was strongly associated with baseline score. Set A ($n=5$) began at 5.00/24 and increased to 10.30 (+5.30), while Set B ($n=16$) began at 10.42 and increased to 11.23 (+0.81). Within TRACE, Set A increased by +4.88 while Set B decreased by -0.46; Control A contained only one participant. These cells are too small and uneven for causal interpretation. The main value of the set analysis is diagnostic: the intended counterbalancing failed in the realised sample, and task difficulty or sample composition could have materially affected apparent pre-post change.

Set	n	Pre /24	Post /24	Change
A	5	5.00 (1.31) [5.00]	10.30 (3.70) [10.00]	+5.30 (3.49) [5.75]
B	16	10.42 (4.59) [10.50]	11.23 (4.67) [10.13]	+0.81 (4.51) [0.50]

Table 5. Combined scores by realised passage set. Group-by-set cells are reported in Supplementary Table S5.

5.5 Sensitivity analysis

Three participants were excluded in the uncertain-authorship sensitivity analysis, leaving TRACE $n=9$ and control $n=9$. Self-review change was +1.25 in TRACE and +0.00 in control

($U=55.0$, $p=.215$); critique change was $+1.19$ and $+1.89$ respectively ($U=32.5$, $p=.507$). Combined change was $+2.44$ in TRACE and $+1.89$ in control ($U=42.0$, $p=.930$). No within-group or between-group test in the corrected sensitivity analysis reached $p<.05$. The sensitivity analysis therefore changes some descriptive magnitudes but not the central conclusion. Detailed task-level and dimension-level sensitivity tables are provided in the supplement.

6. Discussion

6.1 What the active-control comparison does and does not show

The most important result is negative in a useful sense: the pilot does not support a claim that TRACE outperforms another structured reflective framework. The control condition showed a larger mean combined gain in the full sample, although the between-group difference was non-significant and the small sample makes the estimate imprecise. This should not be converted into a claim that control is better. TRACE started substantially higher, the passage sets were badly imbalanced, and the groups had limited headroom at different points on the scale. The defensible inference is narrower: under this design, a TRACE-specific advantage was not detected.

The active comparator is methodologically important because it prevents an easy but weak interpretation of pre-post change. Both groups received prompts that directed attention toward meaning, clarity, language problems and judgement. Both groups also had higher mean post-framework scores. Without a no-framework arm, those increases are compatible with several explanations: a genuine effect of structured reflection, repeated-task familiarity, greater comfort with the rubric-relevant discourse, regression to the mean, or differences in the passages encountered. The current study therefore cannot estimate the causal effect of 'reflection' itself. A future three-arm design is required if that broader claim is to be tested.

Hu et al. (2021) provide a reason to take the broader possibility seriously. Their repeated metacognitive inventory influenced metacognitive awareness and translation quality in student self-training. The present results neither replicate nor contradict that study because the intervention, language pair, training dose and outcomes differ. What this pilot adds is a small active-control comparison and an explicit attempt to measure whether framework-related reasoning appears after prompts are removed. That design choice should be retained and strengthened rather than replaced by an uncontrolled pre-post study.

6.2 Alternatives as a mechanism-specific hypothesis

Alternatives is the clearest place where the data and TRACE's internal logic meet. TRACE explicitly instructs users to generate or compare alternative renderings, and Alternatives was the only rubric dimension on which TRACE showed a larger mean gain than control in the full sample. Borg's (2019) process study supports treating alternative solutions as a genuine component of translatorial decision-making. Yet the observed difference here was tiny relative to uncertainty and was non-significant. Moreover, TRACE began higher on Alternatives. The finding should therefore be used to sharpen the next experiment, not to market TRACE as effective.

A stronger test would pre-specify Alternatives as a mechanism outcome and distinguish quantity from quality. Simply producing more alternatives is not inherently better: alternatives may be implausible, contextually inappropriate or redundant. A revised measure could score whether

alternatives are linguistically plausible, whether they reflect distinct semantic or pragmatic possibilities, and whether the participant can explain the trade-off between them. Such a measure would more directly test what the A step is intended to teach.

6.3 Mnemonic carry-over is not yet retention

The breadth analysis was designed to ask whether a compact mnemonic helps users carry a wider set of reflective questions into an unguided task. TRACE showed a larger descriptive breadth increase, but the comparison was non-significant and the measure was near ceiling from the outset. More fundamentally, the post-task occurred immediately after guided exposure. The study therefore measures immediate unguided carry-over, not memory retention in the ordinary educational sense. Calling the result 'retention' would overstate the temporal evidence.

A future retention design should remove both the written scaffold and the mnemonic for several days, then use novel passages and ask participants to work without reminders. Free recall of the five TRACE elements could be measured separately from performance. This would distinguish remembering the acronym from applying the underlying reasoning. It would also test whether a mnemonic offers any advantage over a structured checklist whose principles may be remembered without a memorable label.

6.4 The pilot is also a measurement study

The reliability results are not a side issue. A framework can appear effective or ineffective depending on how its outcome is operationalised. The low exact category agreement indicates that raters frequently disagreed about whether a response merited 0, 1 or 2 on a specific dimension. Averaging four ratings increased reliability of the total score, but $ICC(A,4)=.635$ is not strong enough to treat the rubric as a mature measurement instrument. Han et al. (2024) show why such disagreement is plausible: rater-mediated interpreting assessment requires attention to source information, scale descriptors and performance features, and raters employ different strategies.

The next version of the rubric should therefore be developed independently of TRACE efficacy testing. Behavioural anchors need to specify observable evidence rather than abstract quality labels; calibration examples should include borderline cases; raters should be trained before scoring; and reliability should be evaluated on a new set of responses. Validity also requires more than agreement. Scores should relate in expected ways to independent indicators of translation expertise or performance, and the rubric should discriminate strong from weak reasoning without simply rewarding verbosity. Until then, 'translation literacy' should be presented as an operational research construct.

6.5 Relevance to asylum-interpreter education

The asylum context supplies a plausible application problem but not an outcome claim. Home Office guidance requires interpreters to identify language or dialect problems, retain all information, avoid unsupported summarisation, request clarification and seek the closest equivalent where culturally difficult language has no straightforward rendering (Home Office, 2025). These requirements entail decisions about context, range, alternatives and consequences. TRACE packages similar questions into a pedagogical sequence, which makes it conceptually relevant to training and reflective debriefing.

That conceptual fit should not be mistaken for ecological validity. The present participants were Arabic-English bilingual adults, not a defined cohort of practising asylum interpreters. They

worked on written constructed passages rather than live dialogue. They did not manage turn-taking, note-taking, working memory, accent, emotion, vulnerable claimants or institutional time pressure. Morrison et al. (2024) demonstrate that emotionally evocative material can reduce interpreting accuracy even in a controlled mock asylum setting. A framework that is usable during an untimed written task may become cognitively costly during live interpreting.

This suggests two different educational uses worth separating. First, TRACE could be tested as a deliberate-practice or post-task debrief tool, where there is time to examine a decision after performance. Second, a shorter internalised version could be tested during simulation to see whether trained interpreters can recognise uncertainty quickly enough to request clarification or flag a closest-equivalent problem without disrupting the interaction. The present study speaks more directly to the first use than the second.

6.6 Qualification, professional standards and reflective scaffolds

The DPSI comparison likewise requires restraint. CIOL's qualification assesses whether candidates can perform defined public-service interpreting and translation tasks; TRACE has not been validated as an assessment of professional competence. A reflective scaffold might eventually support preparation, feedback or continuing professional development, but it should not be used to certify readiness or replace accreditation. Testing with DPSI-qualified interpreters would be informative precisely because it would show whether the framework adds anything once substantial professional competence is already present.

The same distinction applies to structural policy. ICIBI findings about procurement, dialect availability, training and quality assurance cannot be solved by asking individual interpreters to reflect harder. A framework focused on individual cognition should not shift responsibility away from institutions. If TRACE is developed further, its appropriate claim is educational and local: it may help make difficult translation decisions discussable in training. System-level quality still depends on recruitment, language matching, workload, supervision, professional standards and monitoring.

6.7 Robustness and heterogeneity

The uncertain-authorship sensitivity analysis did not overturn the main conclusion. Removing three participants changed the descriptive direction of the combined TRACE-control contrast but still produced no statistically significant group difference. This is useful insofar as the conclusion of 'no detected TRACE-specific advantage' does not rest entirely on those responses. The flagging procedure itself was subjective, however, and should not be treated as a validated AI-detection method.

Participant-level trajectories also show why a larger study matters. Some participants improved sharply, some changed little and others declined. Such heterogeneity may reflect measurement noise, passage difficulty or genuine differences in how participants respond to structured reflection. The present sample is too small to identify moderators. Future studies could pre-specify baseline translation experience, professional status and initial score as candidate moderators, rather than exploring many subgroup effects after data collection.

7. Limitations and priorities for the next study

First, the sample was small. With 21 participants, non-significant tests provide little evidence about equivalence or small effects, and uncertainty around group differences is substantial. The sample also prevents stable moderator analysis. A larger study should be powered around one

pre-specified primary contrast and should report effect estimates and confidence intervals in addition to p-values.

Second, the design lacked a minimal or no-framework comparator. The active control is a strength for testing TRACE specificity, but it means the study cannot determine whether either framework caused the overall pre-post increase. A three-arm design—TRACE, active structured control and repeated assessment without reflective training—would distinguish the general effect of scaffolding from any additional TRACE-specific effect.

Third, randomisation and passage balancing did not survive intact in the analysed sample. The intended block-of-four procedure should have balanced condition and passage version, yet the realised A/B split was 5/16 and Set A was heavily TRACE-weighted. The cause is not recoverable from retained records. Because A and B also differed greatly at baseline, passage difficulty, composition and headroom are entangled with framework effects. Future recruitment should audit allocation logs in real time, stop recruitment if balance fails, and pilot passages for approximate equivalence before the intervention study begins.

Fourth, the participant population and task format limit generalisation. Bilingual Prolific participants are not equivalent to professional interpreters, and self-rated language and translation ability were not independently verified. The written, constructed tasks excluded live speech, memory load, note-taking, interactional repair, emotional testimony and institutional consequences. Future work should first replicate the task with practising interpreters and then progress to spoken simulation before any field use in asylum settings is contemplated.

Fifth, the outcome measure is preliminary. Exact category agreement was low, the averaged ICC was moderate, the original author was one of four raters, and the breadth measure showed a ceiling effect. Although the three additional raters were blinded, rater aggregation cannot compensate for unclear construct boundaries. Rubric development, rater calibration and validation against independent criteria should precede a definitive efficacy trial.

Sixth, the analysis was exploratory and involved multiple outcomes without multiplicity adjustment. Alternatives and breadth were examined because they are theoretically relevant to TRACE, but the present findings should not be treated as confirmatory. A larger study should register its hypotheses, primary outcome and analysis plan in advance, with secondary mechanism analyses clearly labelled.

Seventh, the study measured only immediate carry-over. No delayed assessment tested whether the mnemonic or reasoning survived beyond the session. A delayed unguided test using novel material is essential if TRACE is to be described as memorable or useful for retention.

Eighth, the uncertain-authorship sensitivity analysis depended on researcher judgement. Within-participant discontinuity can have many explanations and does not prove AI assistance. Future online studies should reduce the need for retrospective classification by designing task controls prospectively and by pre-specifying how incomplete, duplicated or otherwise questionable responses will be handled.

Finally, the study was conducted independently without formal institutional ethics review. Adult informed consent, anonymisation and constructed non-case material reduced some risks, but absence of committee review should be reported transparently rather than treated as equivalent to an exemption. Future work involving practising interpreters, simulated trauma or authentic

asylum material should be conducted within an institutional ethics framework appropriate to the setting.

8. Conclusion

This pilot does not demonstrate that TRACE is superior to another structured framework, and it cannot establish that structured reflection caused the observed pre-post increases. Its contribution is more modest and, at this stage, more useful: it operationalises a set of reflective translation behaviours, tests a mnemonic scaffold against an active comparator, exposes important measurement and allocation problems, and identifies mechanism-level hypotheses that can be tested prospectively. Alternatives is the clearest TRACE-aligned candidate outcome, while breadth requires a less ceiling-prone measure and delayed testing.

For asylum-interpreter education, TRACE should therefore be viewed as a candidate reflective scaffold rather than a professional standard or intervention with demonstrated efficacy. A credible next study would use a larger pre-registered sample, balanced and pilot-tested passages, an active and minimal comparator, independently validated scoring, delayed unguided assessment, and practising interpreters completing spoken tasks. Only then can the question move from whether TRACE is a coherent way to structure reflection to whether it improves performance under the demands of high-stakes interpreting.

Statements

Data availability. De-identified aggregate data, scoring documentation and derived analysis materials can be made available by the author to the extent permitted by participant consent. Raw free-text responses are not deposited publicly where participant consent does not permit sharing.

Funding. No external funding was received for this study.

Competing interests. The author declares no competing interests.

Generative AI disclosure. Generative AI was used to create the constructed study passages, as described in the Method, and for language-editing assistance during manuscript preparation. The author retains responsibility for the study design, analysis, interpretation, source checking and final submitted text.

References

- Baker, M. (2018). *In other words: A coursebook on translation* (3rd ed.). Routledge.
- Borg, C. (2019). Written alternative translation solutions in the translation process: A case study. *Translation, Cognition & Behavior*, 2(1), 55–78. <https://doi.org/10.1075/tcb.00019.bor>
- Chartered Institute of Linguists. (n.d.). DPSI overview. <https://www.ciol.org.uk/dpsi-overview>
- Geiling, A., Böttche, M., Knaevelsrud, C., & Stammel, N. (2022a). A comparison of interpreters' wellbeing and work-related characteristics in the care of refugees across different work settings. *BMC Public Health*, 22, 1635. <https://doi.org/10.1186/s12889-022-14034-7>
- Geiling, A., Knaevelsrud, C., Böttche, M., & Stammel, N. (2022b). Psychological distress, exhaustion, and work-related correlates among interpreters working in refugee care: Results of a nationwide online survey in Germany. *European Journal of Psychotraumatology*, 13(1), 2046954. <https://doi.org/10.1080/20008198.2022.2046954>
- Han, C., Zheng, B., Xie, M., & Chen, S. (2024). Raters' scoring process in assessment of interpreting: An empirical study based on eye tracking and retrospective verbalisation. *The*

- Interpreter and Translator Trainer, 18(3), 400–422.
<https://doi.org/10.1080/1750399X.2024.2326400>
- Home Office. (n.d.). Immigration Rules part 11: Asylum (paragraph 339ND). GOV.UK.
<https://www.gov.uk/guidance/immigration-rules/immigration-rules-part-11-asylum>
- Home Office. (2025). Asylum interviews (Version 10.0): Caseworker guidance. GOV.UK.
<https://www.gov.uk/government/publications/conducting-the-asylum-interview-process>
- Hu, Z., Zheng, B., & Wang, X. (2021). The impact of a metacognitive self-regulation inventory in translator self-training: A pre-post study with English-Chinese translation students. *The Interpreter and Translator Trainer*, 15(4), 430–449.
<https://doi.org/10.1080/1750399X.2020.1844436>
- Independent Chief Inspector of Borders and Immigration. (2020). An inspection of the Home Office's use of language services in the asylum process. GOV.UK.
<https://www.gov.uk/government/publications/an-inspection-of-the-home-offices-use-of-language-services-in-the-asylum-process>
- Independent Chief Inspector of Borders and Immigration. (2024). An inspection of asylum casework (June–October 2023). GOV.UK.
<https://www.gov.uk/government/publications/an-inspection-of-asylum-casework-june-october-2023>
- Morrison, L., Given-Wilson, Z., & Memon, A. (2024). The impact of emotionally evocative information on interpreting accuracy in a mock asylum interview. *Applied Cognitive Psychology*, 38(2), e4185. <https://doi.org/10.1002/acp.4185>
- PACTE. (2005). Investigating translation competence: Conceptual and methodological issues. *Meta*, 50(2), 609–619. <https://doi.org/10.7202/011004ar>
- UK Parliament. (2025). Amendment 79: Provision of interpreters and translators under the Immigration Rules.
<https://bills.parliament.uk/bills/3929/stages/20214/amendments/10028010>
- Williamson, A. J. (2024). Defining the role of the UK asylum interpreter: Expectations, realities and training needs [MPhil thesis, University of Glasgow]. Enlighten Theses.
<https://doi.org/10.5525/gla.thesis.84082>